Formal Verification of BLR PCP

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Chapter 1

BLR for general finite fields

1.1 Preliminaries

Let $q = p^s$ for some prime p and $s \in \mathbb{N}$. Let $f : \mathbb{F}^n \to \mathbb{F}_q$ be a function over the finite field $\mathbb{F}_q$. Let $\omega_p = e^{2\pi i/p}$ be the root of unity modolu p.

Definition 1.1.1 (Trace function in finite fields)**.** *The trace function* $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ *is defined as follows:*

$$\mathrm{Tr}(x) = \sum_{i=0}^{s-1} x^{p^i}$$

Definition 1.1.2. *The character group of* $\mathbb{F}_q$*, denoted* $\hat{G}$ *, contains all homomorphisms* $\chi : \mathbb{F}^n \to \mathbb{C}$*.*

Fact 1.1.3. $\hat{G}$ *consists of all the functions:* $\chi_\alpha(x) = \omega_p^{\mathrm{Tr}(\langle \alpha, x \rangle)}$*, for all* $\alpha \in \mathbb{F}^n$*.*

Fact 1.1.4. $\{\chi_\alpha\}_{\alpha \in \mathbb{F}^n}$ *form an orthonormal basis for all functions* $f : \mathbb{F}_q^n \to \mathbb{C}$ *under the inner-product* $\langle f, g \rangle = \mathbb{E}_{x \in \mathbb{F}^n}[f(x)\overline{g(x)}]$*.*

Definition 1.1.5. *Every function* $h : \mathbb{F}^n \to \mathbb{C}$ *has a Fourier expansion* $\hat{h} : \mathbb{F}^n \to \mathbb{C}$ *given by*

$$h(x) = \sum_{\alpha \in \mathbb{F}_q} \hat{h}(\alpha)\chi_\alpha(x)$$

$$\hat{h}(\alpha) = q^{-n} \sum_{x \in \mathbb{F}_q} h(x)\overline{\chi_\alpha(x)}$$

Definition 1.1.6. *The norm of a function* $h : \mathbb{F}^n \to \mathbb{C}$ *is equal to:*

$$\|h\|^2 = \langle h, h \rangle = \mathbb{E}_{x \in \mathbb{F}^n}[h(x)\overline{h(x)}] = q^{-n} \sum_{x \in \mathbb{F}^n} h(x)^2$$

Corollary 1.1.7. *Let* $h : \mathbb{F}^n \to \mathbb{C}$*. Then:* $\|h\| \leq 1$*.*

Proof.

$$\|h\|^2 = q^{-n} \sum_{x \in \mathbb{F}^n} h(x)^2 \leq q^{-n} \sum_{x \in \mathbb{F}^n} |h(x)|^2 \leq 1$$

□

Theorem 1.1.8. *Let $h_1, h_2 : \mathbb{F}^n \to \mathbb{C}$.*

- *Parseval's Identity:* $\|h_1\|^2 = \sum_{\alpha \in \mathbb{F}^n} \widehat{h_1}(\alpha)^2$.
- *Plancheral's Identity:* $\langle h_1, h_2 \rangle = \sum_{\alpha \in \mathbb{F}^n} \widehat{h_1}(\alpha)\widehat{h_2}(\alpha)$.

As there is no isomorphism between $\mathbb{F}_q$ and $\mathbb{C}$, we instead consider $q-1$ functions $\varphi_c : \mathbb{F}^n \to \mathbb{C}$ for all $c \in \mathbb{F}_q^*$:

$$\varphi_c(x) := \omega_p^{\mathrm{Tr}(c \cdot f(x))} = \sum_{\alpha \in \mathbb{F}^n} \widehat{(\varphi_c)}(\alpha)\chi_\alpha(x) \tag{1.1}$$

Definition 1.1.9. *For any $f : \mathbb{F}^n \to \mathbb{F}_q$ we defined the $\mathbb{F}_p$-expansion of f as the set: $\{\varphi_c\}_{c \in \mathbb{F}_q^*}$.*

Definition 1.1.10. *Let $f, g : \mathbb{F}^n \to \mathbb{F}_q$. We defined the distance between f, g as:*

$$\delta(f, g) = \Pr_{x \in \mathbb{F}^n}[f(x) \neq g(x)]$$

Now, we express the distance of f, g in terms of the Fourier coefficients $\mathbb{F}_p$-expansion sets.

Claim 1.1.11. *Let $f, g : \mathbb{F}^n \to \mathbb{F}_q$ and their $\mathbb{F}_p$-expansion sets be $\{\varphi_c\}_{c \in \mathbb{F}_q^*}, \{\gamma_c\}_{c \in \mathbb{F}_q^*}$ respectively.*

$$\delta(f, g) = 1 - \frac{1}{q}\left(1 + \sum_{c \in \mathbb{F}_q^*} \langle \varphi_c, \gamma_c \rangle\right) = 1 - \frac{1}{q}\left(1 + \sum_{c \in \mathbb{F}_q^*} \sum_{\alpha \in \mathbb{F}^n} \widehat{\varphi_c}(\alpha)\overline{\widehat{\gamma_c}(\alpha)}\right)$$

Definition 1.1.12. *Let $\mathcal{LIN}$ be the set of linear functions form $\mathbb{F}^n$ to $\mathbb{F}_q$. For $\alpha \in \mathbb{F}^n$, we denote $\ell_\alpha(x) = \langle x, \alpha \rangle$. Then:*

$$\mathcal{LIN} = \{\ell_\alpha\}_{\alpha \in \mathbb{F}^n}$$

Claim 1.1.13. *Let $f : \mathbb{F}^n \to \mathbb{F}_q$ with $\mathbb{F}_p$-expansion sets be $\{\varphi_c\}_{c \in \mathbb{F}_q^*}$ and let $\ell_\alpha \in \mathcal{LIN}$ for some $\alpha \in \mathbb{F}^n$, and let it's $\mathbb{F}_p$-expansion set be $\{\lambda_c\}_{c \in \mathbb{F}_q^*}$. Then:*

$$\delta(f, \ell_\alpha) = 1 - \left(\frac{1}{q}\left(1 + \sum_{c \in \mathbb{F}_q^*} \widehat{\varphi_c}(c\alpha)\right)\right)$$

Proof. For all $\beta \in \mathbb{F}^n$, the following is 0 if $\beta \neq c\alpha$ and 1 otherwise:

$$\widehat{\lambda_c}(\beta) = q^{-n} \sum_{x \in \mathbb{F}^n} \lambda_c(x)\overline{\chi_\beta(x)} = q^{-n} \sum_{x \in \mathbb{F}^n} \omega_p^{\mathrm{Tr}(c \cdot \ell_\alpha(x)) - \mathrm{Tr}(\langle \beta, x \rangle)} = q^{-n} \sum_{x \in \mathbb{F}^n} \omega_p^{\mathrm{Tr}(\langle c\alpha, x \rangle) - \mathrm{Tr}(\langle \beta, x \rangle)}$$

□

Corollary 1.1.14. *For all $\alpha \in \mathbb{F}^n$, $-1 \leq \sum_{c \in \mathbb{F}_q^*} \widehat{\varphi_c}(c\alpha) \leq q - 1$ is a real number.*

Proof. This follows because $\delta(f, \ell_\alpha) \in [0, 1]$. □

Proof. Equivalent to $0 \leq \delta \leq 1 - 1/q$ □

Definition 1.1.15. *The distance of $f : \mathbb{F}^n \to \mathbb{F}_q$ from $\mathcal{LIN}$is defined as: $\delta(f, \mathcal{LIN}) := \min_{\alpha \in \mathbb{F}^n} \delta(f, \ell_\alpha)$.*

Corollary 1.1.16.

$$\delta(f, \mathcal{LIN}) = \min_{\alpha \in \mathbb{F}^n} \left(1 - \left(\frac{1}{q}\left(1 + \sum_{c \in \mathbb{F}_q^*} \widehat{\varphi_c}(c\alpha)\right)\right)\right)$$

Equivalently,

$$\delta(f, \mathcal{LIN}) = 1 - \frac{1}{q}\left(1 + \max_{\alpha \in \mathbb{F}^n}\left(\sum_{c \in \mathbb{F}_q^*} \widehat{\varphi_c}(c\alpha)\right)\right)$$

1.2 BLR Test over Finite Fields

The BLR Test Given oracle access to a function $f : \mathbb{F}^n \to \mathbb{F}_q$, the verifier performs the following:

- Sample random $a, b \in \mathbb{F}^n \setminus \{0\}$ and sample $x, y \in \mathbb{F}^n$.
- $\mathsf{V}^f_{\mathsf{BLR}} = 1 \iff a \cdot f(x) + b \cdot f(y) = f(a \cdot x + b \cdot y)$.

Theorem 1.2.1. *Let $f : \mathbb{F}^n \to \mathbb{F}_q$ with $\mathbb{F}_p$-expansion sets be $\{\varphi_c\}_{c \in \mathbb{F}_q^*}$.*

$$\Pr_{a,b,x,y}\left[\mathsf{V}^f_{\mathsf{BLR}} = 1\right] = \frac{1}{q}\left(1 + \frac{1}{(q-1)^2}\sum_{\alpha \in \mathbb{F}^n}\left(\sum_{a \in \mathbb{F}_q^*} \widehat{\varphi_a}(a\alpha)\right)^3\right)$$

Proof. Let the $\mathbb{F}_p$-expansion set of f be $\{\varphi_c\}_{c \in \mathbb{F}_q^*}$. Let also $a, b \in \mathbb{F}^n \setminus \{0\}$ and $x, y \in \mathbb{F}^n$. We observe that:

$$\mathsf{V}^f_{\mathsf{BLR}} = 1 \iff \frac{1}{q}\left(\sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by)}\right) = 1$$

As the above sum is zero unless $af(x) + bf(y) - f(ax + by) = 0$. Hence:

$$\begin{aligned}\Pr_{a,b,x,y}\left[\mathsf{V}^f_{\mathsf{BLR}} = 1\right] &= \mathbb{E}_{a,b,x,y}\left[\frac{1}{q}\left(\sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by))}\right)\right] \\ &= \frac{1}{q} + \mathbb{E}_{a,b,x,y}\left[\sum_{c \in \mathbb{F}_q^*} \omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by))}\right]\end{aligned}$$

Now, we consider the last term:

$$
\begin{aligned}
&\mathbb{E}_{a,b,x,y}\left[\sum_{c\in\mathbb{F}_q^*}\omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by))}\right]\\
&=\mathbb{E}_{a,b,x,y}\left[\sum_{c\in\mathbb{F}_q^*}\varphi_{ca}(x)\varphi_{cb}(y)\varphi_{-c}(a\cdot x+b\cdot y)\right]\\
&=\mathbb{E}_{a,b,x,y}\left[\sum_{c\in\mathbb{F}_q^*}\sum_{\alpha,\beta,\gamma\in\mathbb{F}^n}\widehat{\varphi_{ca}}(\alpha)\chi_\alpha(x)\widehat{\varphi_{cb}}(\beta)\chi_\beta(y)\widehat{\varphi_{-c}}(\gamma)\chi_\gamma(ax+by)\right]\\
&=\mathbb{E}_{a,b,x,y}\left[\sum_{c\in\mathbb{F}_q^*}\sum_{\alpha,\beta,\gamma\in\mathbb{F}^n}\widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(\beta)\widehat{\varphi_{-c}}(\gamma)\chi_\alpha(x)\chi_\beta(y)\chi_\gamma(ax+by)\right]\\
&=\mathbb{E}_{a,b,x,y}\left[\sum_{c\in\mathbb{F}_q^*}\sum_{\alpha,\beta,\gamma\in\mathbb{F}^n}\widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(\beta)\widehat{\varphi_{-c}}(\gamma)\chi_\alpha(x)\chi_\beta(y)\chi_{a\gamma}(x)\chi_{b\gamma}(y)\right]\\
&=\mathbb{E}_{a,b}\left[\sum_{c\in\mathbb{F}_q^*}\sum_{\alpha,\beta,\gamma\in\mathbb{F}^n}\widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(\beta)\widehat{\varphi_{-c}}(\gamma)\mathbb{E}_{x,y}\left[\chi_\alpha(x)\chi_\beta(y)\chi_{a\gamma}(x)\chi_{b\gamma}(y)\right]\right]
\end{aligned}
$$

Now, we consider the following expression:

$$
\mathbb{E}_{x,y}\left[\chi_\alpha(x)\chi_\beta(y)\chi_{a\gamma}(x)\chi_{b\gamma}(y)\right]=\mathbb{E}_{x,y}\left[\omega_p^{\mathrm{Tr}(\langle\alpha,x\rangle+\langle\beta,y\rangle+\langle\gamma,ax\rangle+\langle\gamma,by\rangle)}\right]
$$

Observe that the above is zero, unless $\gamma=-a^{-1}\alpha$ and $\beta=a^{-1}b\alpha$. Thus:

$$
\begin{aligned}
&\mathbb{E}_{a,b,x,y}\left[\sum_{c\in\mathbb{F}_q^*}\omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by))}\right]\\
&=\mathbb{E}_{a,b}\left[\sum_{c\in\mathbb{F}_q^*}\sum_{\alpha,\beta,\gamma\in\mathbb{F}^n}\widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(\beta)\widehat{\varphi_{-c}}(\gamma)\mathbb{E}_{x,y}\left[\chi_\alpha(x)\chi_\beta(y)\chi_{a\gamma}(x)\chi_{b\gamma}(y)\right]\right]\\
&=\mathbb{E}_{a,b}\left[\sum_{c\in\mathbb{F}_q^*}\sum_{\alpha\in\mathbb{F}^n}\widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(a^{-1}b\alpha)\widehat{\varphi_{-c}}(-a^{-1}\alpha)\right]\\
&=\mathbb{E}_{a,b}\left[\sum_{c\in\mathbb{F}_q^*}\sum_{\alpha\in\mathbb{F}^n}\widehat{\varphi_{ca}}(ca\alpha)\widehat{\varphi_{cb}}(cb\alpha)\widehat{\varphi_{-c}}(-c\alpha)\right]\\
&=\mathbb{E}_{a,b}\left[\sum_{c\in\mathbb{F}_q^*}\sum_{\alpha\in\mathbb{F}^n}\widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha)\widehat{\varphi_c}(c\alpha)\right]\\
&=\frac{1}{(q-1)^2}\sum_{a,b,c\in\mathbb{F}_q^*}\sum_{\alpha\in\mathbb{F}^n}\widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha)\widehat{\varphi_c}(c\alpha)\\
&=\frac{1}{(q-1)^2}\sum_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)^3
\end{aligned}
$$

□

Theorem 1.2.2. *Let $f : \mathbb{F}^n \to \mathbb{F}_q$.*

$$(1-\delta(f,\mathcal{LIN}))^3 \leq \Pr_{a,b,x,y}\left[\mathsf{V}^f_{\mathsf{BLR}} = 1\right] \leq 1-\delta(f,\mathcal{LIN})$$

Proof. For the left hand-side:

$$\begin{aligned}
\Pr\left[\mathsf{V}^f_{\mathsf{BLR}} = 1\right] &= \frac{1}{q}\left(1+\frac{1}{(q-1)^2}\sum_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)^3\right) \\
&\leq \frac{1}{q}\left(1+\frac{1}{(q-1)^2}\max_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)\sum_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)^2\right) \\
&= \frac{1}{q}\left(1+\frac{1}{(q-1)^2}\max_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)\sum_{\alpha\in\mathbb{F}^n}\left(\sum_{a,b\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha)\right)\right) \\
&= \frac{1}{q}\left(1+\frac{1}{(q-1)^2}\max_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)\sum_{a,b\in\mathbb{F}_q^*}\left(\sum_{\alpha\in\mathbb{F}^n}\widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha)\right)\right)
\end{aligned}$$

Consider the following, due to Plancherel's identity and the fact that φ_a map to the roots of unity modulo p:

$$\sum_{\alpha\in\mathbb{F}^n}\widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha) = \mathbb{E}_\alpha\left[\varphi_a(a\alpha)\gamma_b(b\alpha)\right] \leq \mathbb{E}_\alpha\left[|\varphi_a(a\alpha)\gamma_b(b\alpha)|\right] \leq 1$$

Hence, it follows that:

$$\Pr_{a,b,x,y}\left[\mathsf{V}^f_{\mathsf{BLR}} = 1\right] \leq \frac{1}{q}\left(1+\max_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)\right) = 1-\delta(f,\mathcal{LIN})$$

For the left hand side of the claim:

$$\begin{aligned}
\Pr\left[\mathsf{V}^f_{\mathsf{BLR}} = 1\right] &= \frac{1}{q}\left(1+\frac{1}{(q-1)^2}\sum_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)^3\right) \\
&\geq \frac{1}{q}\left(1+\frac{1}{(q-1)^2}\max_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)^3\right) \\
&\geq \left(\frac{1}{q}\left(1+\max_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)\right)\right)^3 \\
&= (1-\delta(f,\mathcal{LIN}))^3
\end{aligned}$$

where the last inequality follows from the following: for every $q \geq 2, A \geq \frac{1-|\mathbb{F}|^2}{2q-1}$:

$$\left(\frac{1}{q}(1+A)\right)^3 \leq \frac{1}{q}\left(1+\frac{1}{(q-1)^2}A^3\right)$$

(we plug $A = \max_{\alpha\in\mathbb{F}^n}\left(\sum_{a\in\mathbb{F}_q^*}\widehat{\varphi_a}(a\alpha)\right)$ in the expression above. By **??**, we observe that the above holds for any q).

□

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Definition 1.3.1. *A function $f : F^n \to F$ is linear if there exists $a \in F^n$ such that*

$$f(x) = \sum_i a_i x_i$$

for every $x \in F^n$.

Definition 1.3.2. *The finite-field BLR test samples $x, y \in F^n$ and $a, b \in F \setminus \{0\}$, and accepts iff*

$$af(x) + bf(y) = f(ax + by).$$

Definition 1.3.3. *The rejection probability of the finite-field BLR test is*

$$\Pr[V_{BLR}^f = 0] = 1 - \Pr[V_{BLR}^f = 1].$$

Definition 1.3.4. *Write $F = \mathbb{F}_q$ where $q = p^m$ for a prime number p and a positive integer m. For each $\alpha \in \mathbb{F}_q^n$, define the additive character $\chi_\alpha : \mathbb{F}_q \to \mathbb{C}$ as*

$$\chi_\alpha(x) := \omega_p^{\mathrm{Tr}(\langle\alpha,x\rangle)},$$

where $\omega_p = \exp(2i/p)$ and $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ is the field trace $\mathrm{Tr}(a) = \sum_{j=0}^{m-1} a^{p^j}$.

Lemma 1.3.5. *The family $\{\chi_\alpha\}_{\alpha\in\mathbb{F}_q^n}$ is an orthonormal basis for the complex-valued functions $g : \mathbb{F}_q^n \to \mathbb{C}$ with respect to the inner product*

$$\langle g, h\rangle := \mathbb{E}_{x\in\mathbb{F}_q^n}[g(x)\overline{h(x)}].$$

Lemma 1.3.6. *Every function $g : \mathbb{F}_q^n \to \mathbb{C}$ has a unique Fourier expansion*

$$g(x) = \sum_{\alpha\in\mathbb{F}_q^n} \widehat{g}(\alpha)\chi_\alpha(x).$$

Lemma 1.3.7. *Parseval and Plancherel hold:*

$$\langle g, g\rangle = \sum_\alpha |\widehat{g}(\alpha)|^2, \qquad \langle g, h\rangle = \sum_\alpha \widehat{g}(\alpha)\overline{\widehat{h}(\alpha)}.$$

Definition 1.3.8. *For $f : \mathbb{F}_q^n \to \mathbb{F}_q$, define its Fourier set*

$$\Phi(f) := \{\varphi_c\}_{c\in\mathbb{F}_q^\times}, \qquad \varphi_c(x) := \omega_p^{\mathrm{Tr}(cf(x))}.$$

Lemma 1.3.9. *If $f, g : \mathbb{F}_q^n \to \mathbb{F}_q$ have Fourier sets $\{\varphi_c\}_{c \in \mathbb{F}_q^\times}$ and $\{\psi_c\}_{c \in \mathbb{F}_q^\times}$, then*

$$\Delta(f, g) = 1 - \frac{1}{q}\left(1 + \sum_{c \in \mathbb{F}_q^\times} \langle \varphi_c, \psi_c \rangle\right).$$

Lemma 1.3.10. *If f has Fourier set $\{\varphi_c\}_{c \in \mathbb{F}_q^\times}$, then its distance to the linear functions is*

$$\Delta(f, \mathrm{LIN}) = 1 - \frac{1}{q}\left(1 + \max_{\alpha \in \mathbb{F}_q^n} \sum_{c \in \mathbb{F}_q^\times} \widehat{\varphi_c}(c\alpha)\right).$$

Lemma 1.3.11. *A Fourier-analytic calculation shows that the rejection probability of the finite-field BLR test dominates the Fourier-set expression for $\Delta(f, \mathrm{LIN})$*

Lemma 1.3.12. *For every function $f : F^n \to F$, the finite-field BLR test satisfies*

$$\Pr[V_{BLR}^f = 0] \geq \Delta(f, \mathrm{LIN}).$$

This is the improved finite-field analysis proved via Fourier analysis.

Lemma 1.3.13. *If the finite-field BLR test accepts f with probability at least $1 - \varepsilon$, then*

$$\Delta(f, \mathrm{LIN}) \leq \varepsilon.$$

Lemma 1.3.14. *When $q = 2$, this reduces to the Boolean Fourier-analytic proof of the BLR theorem.*

Chapter 2

Gowers Hatami theorem

Chapter 3

Linear PCP for linear equations

Chapter 4

Linear PCP for tensor checks

Chapter 5

Linear PCP to PCP